

Some Practice Questions of Derivative**Chapter:0 [Relation between continuity and differentiability]**

1. Write one example of the function which is continuous at but not differentiable at that point.
2. Define derivative of a function at a point.
3. What is the relation between continuity and differentiability of a function at a point?

Chapter:1 Derivative of hyperbolic functions**Knowledge Based Questions:**

1. Define hyperbolic functions.
2. What is the derivative of $\cosh x$, $\sinh x$, $\tanh x$?
3. What is the derivative of, $\cosh^{-1} x$, $\sinh^{-1} x$?
4. Under which necessary condition, $\frac{d}{dx}(\operatorname{sech}^{-1} x) = \frac{-1}{x\sqrt{1-x^2}}$?
5. What is the derivative of $\frac{e^x + e^{-x}}{e^x - e^{-x}}$?

Understanding & Application Based Questions:

1. Find the derivative of :
i) $\sinh x$ ii) $\tanh x$ iii) $\cosh^{-1} x$ iv) $\tanh^{-1} x$
2. Find the derivative of following functions:

Type:1[Hint: use Chain rule]

- i) $\tanh(\sin^{-1} x)$ ii) $2 \tanh^{-1}(\tan \frac{x}{2})$ iii) $\operatorname{arc} \tan \sinh x$ iv) $2 \tanh^{-1}(\tanh \frac{x}{2})$
v) $\sinh^{-1}(\cos 2x)$ vi) $e^{\cosh^{-1} \frac{x}{a}}$ vii) $\log(\cosh x^2)$

Answers

- i) $\frac{\operatorname{sech}^2(\sin^{-1} x)}{\sqrt{1-x^2}}$ ii) $\sec x$ iii) $\operatorname{sech} x$ iv) $\operatorname{sech} x$ v) $\frac{-2 \sin 2x}{\sqrt{1+\cos^2 2x}}$ vi) $\frac{e^{\cosh^{-1} \frac{x}{a}}}{\sqrt{x^2-a^2}}$ vii) $2x \tanh x^2$

Type: 2[Hint: Take log/ ln first then differentiate on both sides]

- i) $(\sinh x)^{\cosh x}$ ii) $(\ln x)^{\sinh x}$ iii) $x^{\sinh^2 \frac{x}{a}}$ iv) $(\cosh x)^{\sinh^{-1} x}$ v) $\left(\sinh \frac{x}{a} + \cosh \frac{x}{a}\right)^{nx}$

Answers:

- i) $(\sinh x)^{\cosh x} (\cosh x \cdot \cot h x + \sinh x \cdot \ln(\sinh x))$ ii) $(\ln x)^{\sinh x} \left(\frac{\sinh x}{x \cdot \ln x} + \cosh x \cdot \ln(\ln x)\right)$
iii) $x^{\sinh^2 \frac{x}{a}} \left(\frac{1}{x} \sinh^2 \frac{x}{a} + \frac{1}{a} \sinh \frac{2x}{a} \cdot nx\right)$ iv) $\cosh x^{\sinh^{-1} x} (\sinh^{-1} x \tanh x + \frac{1}{\sqrt{1+x^2}} \ln \cosh x)$
v) $n \left(\sinh \frac{x}{a} + \cosh \frac{x}{a}\right)^{nx} \left[\frac{x}{a} + \ln \left(\sinh \frac{x}{a} + \cosh \frac{x}{a}\right)\right]$

Chapter:2 (L-Hospital Rule)**Knowledge Based Questions:**

1. Write 7 indeterminate forms
2. State L-Hospital rule.
3. Write any one condition where we use the derivative to find the limit of a function.
4. Is L-Hospital's rule applicable for finding the limiting value of the function in any indeterminate forms?
5. Write two indeterminate in which L-Hospital's rule is applicable?
6. Write a characteristic of L-Hospital rule.

Understanding & Application Based Questions:

7. Find the limit of the following function using L-Hospital's rule.

- a) $\lim_{x \rightarrow 0} \frac{(e^x - 1) \tan x}{x^2}$ b) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x - \sin x}$ c) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2 \cos x}{\sin^2 x}$ d) $\lim_{x \rightarrow 0} \frac{x \cdot e^x - \ln(x+1)}{x^2}$
e) $\lim_{x \rightarrow 0} \frac{x - \sin x}{\tan^3 x}$ f) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x}\right)$ g) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec 3x}{\sec x}$ h) $\lim_{x \rightarrow 0} \frac{2 \cos x - 2 + x^2}{x^4}$
i) $\lim_{x \rightarrow 0} \frac{\ln x^2}{\cot x^2}$ j) $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln x}\right)$ k) $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x}\right)$ l) $\lim_{x \rightarrow 0^+} x^x$

$$\text{m) } \lim_{x \rightarrow 0^+} \sin x \cdot \ln x^2 \quad \text{n) } \lim_{x \rightarrow \infty} \frac{\ln(x^2+1)}{\ln(x^3+1)} \quad \text{o) } \lim_{x \rightarrow \infty} \frac{5x^2+4x-3}{2x^2-3x+5}$$

Answers:

$$\text{a) } 1 \quad \text{b) } 3 \quad \text{c) } 2 \quad \text{d) } 3/2 \quad \text{e) } 1/6 \quad \text{f) } 0 \quad \text{g) } -1/3 \quad \text{h) } 1/12$$

$$\text{i) } 0 \quad \text{j) } 1/2 \quad \text{k) } -1/3 \quad \text{l) } 1 \quad \text{m) } 0 \quad \text{n) } 2/3 \quad \text{o) } 5/2$$

Chapter 3: (Tangent and Normal)**Knowledge Based Questions:**

1. What is the geometrical meaning of $\frac{dy}{dx}$?
2. What is the physical meaning of $\frac{dy}{dx}$?
3. What is the slope of the tangent parallel to x - axis?
4. What is the slope of the tangent parallel to y - axis?
5. What do you understand by angle between two curves?

Understanding & Application Based Questions:

1. Find the equation of tangent and normal to the following curve.
 a. $y = 2x^3 - 5x^2 + 8$ at (2,4) b. $x^2 - y^2 = 7$ at (4, -3) c. $x^2 + 3xy + y^2 = 5$ at (1,1)

Answers:

- (a) $4x - y = 4$; $x + 4y = 18$ (b) $4x + 3y = 7$; $3x - 4y = 24$ (c) $x + y = 2$; $x - y = 0$
2. Show that the tangents to the curves $y = 3x^3$ at the point $x = 2$ and $x = -2$ are parallel.
3. Find the inclination of the tangent with the x-axis to the curve $x^2 + y^2 = 25$ at (5,0). **Ans:** $\frac{\pi}{2}$
4. At what angle does the curve $y(1+x) = x$ cut x-axis? **Ans:** $\frac{\pi}{4}$
5. Find the point on the curve $y = 2x^2 - 6x + 9$ where the tangent is parallel to x-axis. **Ans:** $(\frac{3}{2}, 0)$
6. Find the point on the curve $y = 2x^2 + 3x + 5$, tangent at which is perpendicular to the line $5y - x = 10$. **Ans:** $(-2, 7)$
7. Find the equation of tangent to the curve $x^2 + 3y - 10 = 0$, which is perpendicular to the line $3x + 2y + 15 = 0$. **Ans:** $2x - 3y + 1 = 0$
8. Find the equations of horizontal and vertical tangents to the curve $x^2 + y^2 - xy - 27 = 0$. **Ans:** $y = \pm 6$; $x = \pm 6$
9. Find the point on the curve $y^2 = 4ax$, the tangent at which is inclined to x-axis at 45° . **Ans:** $(a, 2a)$
10. Find the angle of intersection of following curves.
 a. $y = x^2$ and $x = y^2$ b. $4y = x^2 + 12$ and $y^2 = 8x$ at (2,4)
 c. $x^2 + 4y^2 = 8$ and $x^2 - 2y^2 = 4$ d. $xy = 6$ and $x^2y = 12$ e. $y^2 = x^3$ and $y = 2x$ at (0,0)

Answers:

- a) $\frac{\pi}{2}$ at (0,0) and $\tan^{-1}(\frac{3}{4})$ at (1,1) b) 0° c) 0° d) $\tan^{-1}(\frac{3}{11})$ e) $\tan^{-1}(2)$
11. Show that the curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 - 2 = 0$ cut orthogonally.

Chapter 4: (Differential Approximation)**Knowledge Based Questions:**

1. What does dy represent?
2. What does Δy represent?
3. What is the difference between dy and Δy ?
4. Write the formula to find the error in the differential approximation.
5. How do you find the percentage error in the differential approximation?
6. Write the differential of $\sin x$, $\cos x$, $\tan x$?
7. Let, $y = f(x)$ be a function then write the expression for
 a. actual change in y b. approximate change in y c. percentage error

Understanding & Application Based Questions:

8. By using different approximation, find the value of $\sqrt{16.4}$ **Ans:** 4.05
9. Find the approximated increase in the surface area of a cube if each edge increases from 10 to 10.01. Also calculate percentage error in the use of differential approximation. **Ans:** 1.2 cm^2 ; 1.2006 cm^2 ; 0.0006; 0.0001%
10. Find the approximate Volume of a sphere when its radius increases from 6 to 6.3 cm. **Ans:** $331.2\pi \text{ cm}^3$

Chapter 5: (Rate Measure)

Knowledge Based Questions:

1. If 's' is the distance covered by the particle along the straight line in time 't', then what does $\frac{ds}{dt}$ and $\frac{d^2s}{dt^2}$ represent?
2. What is the difference between average rate of change and instantaneous rate of change of y with respect to x?

Application Based Questions:

Type:1

- a) A circular disc of radius 3 cm is being heated. Due to expansion, its radius increases at the rate of 0.05 cm/sec. Find the rate at which its area is increasing when radius is 3.2cm. **Ans:** $0.32\pi \text{ cm}^2/\text{sec}$
- b) The side of an equilateral triangle is increases uniformly at the rate of 2cm/sec. At what rate is the area increasing when the side is 20cm. **Ans:** $20\sqrt{3} \text{ cm}^2/\text{sec}$
- c) The volume of metal in a hollow sphere is constant. If the inner radius is increasing at the rate of 1cm/sec, find the rate of increase of the outer radius when the radii are 4cm and 8 cm respectively. **Ans:** $\frac{1}{4} \text{ cm/sec}$
- d) If the volume of an expanding cube is increasing at the rate of $4 \text{ cm}^3/\text{sec}$, how fast is its surface area increasing when the surface area is 24 cm^2 ? **Ans:** $8 \text{ cm}^2/\text{sec}$
- e) Water is poured into a right circular cylinder of radius 3 cm at the rate of $36 \text{ cm}^3/\text{min}$. Find the rate at which the level of water is rising? **Ans:** $\frac{4}{\pi} \text{ cm/min}$

Type:2

- a) If the volume of cube is increasing at the constant rate, prove that the rate of increase in surface area varies inversely as the length of the edge of the cube.
- b) If the area of the circle is increasing at the uniform rate, prove that the increase of perimeter varies inversely as radius.
- c) A spherical ball of salt is dissolving in water in such a way that the rate of decrease of the volume at any instant is proportional to the surface area. Prove that the radius is decreasing at a constant rate.
- d) A curve $y^2 = 8x$ changes its abscissa and ordinate at the same rate. The ordinate is double of its abscissa. Justify it with calculation.
- e) The rate of change of volume of a sphere is proportional to the square of its diameter. Explain

Type 3:

- a) The radius of a cylinder is increasing at the rate of 2 cm/sec and its altitude is decreasing at the rate of 3 cm/sec. Find the rate of change of volume when radius is 3 cm and altitude 5 cm. **Ans:** $33\pi \text{ cm}^3/\text{sec}$.

Type 4:

- a) Sand is pouring from a pipe at the rate of $12 \text{ cm}^3/\text{s}$. The falling sand forms a cone on the ground in such a way that the height of the cone always one-sixth of the radius of the base. How fast is the height of the sand-cone increasing when the height is 4 cm? **Ans:** $\frac{1}{48\pi} \text{ cm/sec}$
- b) Water flows into an inverted canonical tank at the rate of $42 \text{ ft}^3/\text{min}$. When the depth of the water is 8ft, how fast is the level rising if the height of the tank is 12ft and radius at the top is 6 ft. **Ans:** $\frac{21}{8\pi} \text{ ft/min}$

c) Water is leaking from a conical funnel at the rate of $5 \text{ cm}^3/\text{sec}$. If the radius of the base of the funnel is 5 cm and the altitude is 10 cm, find the rate at which the water level is dropping when it is 2.5 cm from the top.

$$\text{Ans: } \frac{16}{45\pi} \text{ cm/sec}$$

Type : 5

a) A man 2 meters high walks at a uniform speed of 5 km/hour away from a lamp-post 6 meters high. Find the rate of which the length of his shadow increases.

$$\text{Ans: } 2.5 \text{ km/hour}$$

Type:6

b) A ladder 5 m long is leaning against a wall. The bottom of the ladder is pulled along the ground, toward the wall, at the rate of 2 cm/sec. How fast is its height on the wall increasing when the foot of the ladder is 4 m away from the wall?

$$\text{Ans: } \frac{8}{3} \text{ cm/sec}$$

c) A ladder 16 m long is leaning against a wall. The bottom of the ladder is pulled along the ground, away from the wall, at the rate of 2 m/sec. How fast is its height on the wall decreasing when the foot of the ladder is 4 m away from the wall?

$$\text{Ans: } 2/\sqrt{15} \text{ m/s}$$

d) A kite in 120 m high and 130 m of string is out. If the kite is moving away horizontally at the rate of 52 m/sec, find the rate at which the string is being paid out.

$$\text{Ans: } 20 \text{ m/sec}$$

Type: 7

a) A point is moving along the curve $y = 2x^3 - 3x^2$ in such way that its x-co-ordinate is incre at the rate of 4 ft./sec. Find the rate at which the distance of the point from the ori increasing when the point is at (2, 4).

$$\text{Ans: } 20\sqrt{5} \text{ ft./sec}$$

b) A ship leaves a port at mid night and travels east at the rate of 9km/hr. Another ship leaves the same port after 1 hour and sails south at the rate of 12km./hr. how fast are they separating at 2:00A.M?

$$\text{Ans: } \frac{51}{\sqrt{13}} \text{ km/hr}$$

c) Two aeroplanes in flight cross above town at 1 pm. One plane travels east at 300km/hr, while other towards north at 400km/hr. At what rate does the distance between the plane changes at 3:00pm.

$$\text{Ans: } 1000 \text{ km/hr}$$

d) Two cars start from certain places goes same instant. One goes east at 60 km/hr and other them south at 80 km/hr. How fast is the distance between increasing? Express in symbolic form.

$$\text{Ans: } 100 \text{ km/hr}$$

Type-8

Q. Two concentric circles are expanding in such a way that the radius of the inner circle is increasing at the rate of 10 cm/sec and that of the outer circle at the rate of 7 cm/sec. At a certain time, the radii of the inner and the outer circles are respectively 24 cm and 30 cm. At that time, is the area between the circles increasing or decreasing? **Ans: $60\pi \text{ cm}^2/\text{sec}$ (decreasing)**

Some Additional Questions:

1. Water is dripping out from a conical funnel, at the uniform rate of $2 \text{ cm}^3/\text{sec}$ through a tiny hole at the vertex at the bottom. When the slant height of the water is 4 cm, find the rate of decrease of the slant height of the water given that the vertical angle of the funnel is 120° .

$$\text{Ans: } \frac{1}{3\pi} \text{ cm/sec}$$

2. A man is moving away from a tower 85 ft high at a speed of 4 ft/sec. Find the rate at which his angle of elevation of the top of the tower is changing, when he is at a distance of 60 ft from the foot of the tower. Assume the eye-level of man to be 5 ft from the ground.

$$\text{Ans: } -0.032 \text{ rad/sec}$$

3. The area of an expanding rectangle is increasing at the rate of $48 \text{ cm}^2/\text{sec}$. The length of the rectangle is always equal to the square of the breadth. At what rate the length is increasing at the instant when the breadth is 4.5 cm?

$$\text{Ans } 7.11 \text{ cm/sec}$$

4. The length x of a rectangle is decreasing at the rate of 5 cm/minute and the width y is increasing at the rate of 4 cm/minute. When $x = 8 \text{ cm}$ and $y = 6 \text{ cm}$, find the rate of change of

(a) the perimeter and

(b) the area of the rectangle.

Ans: $-2 \frac{\text{cm}}{\text{minute}}$; $2 \frac{\text{cm}^2}{\text{minute}}$

5. A vessel is in the form of an inverted cone of semi vertical angle 45° . Water is poured into this vessel at the rate of 100 cc per second. Find the rate of rise in the water level when it is 2 cm deep. Ans: $\frac{25}{\pi} \text{ cm/sec}$

